ETCS Writing Toolkit

Engineering, Technology & Computer Science Communication

ENGL 23401 Technical Report Writing | ENGL 42101 Technical Writing Projects

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This toolkit is designed to help Engineering, Technology, and Computer Science (ETCS) students produce professional-quality technical documents. Each component includes a template, annotated guidance, and examples aligned with real-world engineering communication standards. Use the components in sequence or independently as your assignment requires.

What's in This Toolkit

- Component 1: Tone & Style Guidelines for Technical Writing
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> **HOW TO USE THIS TOOLKIT:** Templates show you structure. Annotations (marked with 📌) explain the why behind each section. Read the annotations before you fill in the template; they will save you time and help you make better decisions.

Component 1

Tone & Style Guidelines

Used in: ENGL 23401 and ENGL 42101

Technical writing is not academic writing. It is not casual writing. It occupies a specific register that engineers and technical professionals use to communicate decisions, findings, and recommendations clearly and credibly.

The Four Principles of Technical Tone

1. Clarity Over Complexity

Technical writing prioritizes precision and accessibility. Avoid jargon unless your audience shares it. Define terms on first use. Use the simplest word that is accurate.

> **DO THIS:** The sensor failed to transmit data when ambient temperature exceeded 40°C.

> **NOT THIS:** The aforementioned transducer device exhibited transmission irregularities in elevated thermal environmental conditions.

> 📌 **ANNOTATION:** *Plain language is not less professional. It is more professional. Readers who have to work to understand your writing will not trust your analysis.*

2. Active Voice, Specific Subjects

Technical writing should tell the reader who did what. Passive voice hides responsibility and creates ambiguity.

> **DO THIS:** The team tested three material samples under load conditions.

> **NOT THIS:** Three material samples were tested under load conditions.

> 📌 **ANNOTATION:** *Exception: passive voice is acceptable when the actor is unknown, irrelevant, or when emphasizing the object of the action is standard practice in your field (e.g., 'The circuit was grounded').*

3. Audience Awareness

Always identify your primary and secondary audiences before you write. Technical writing often has layered audiences: a supervisor who needs the decision, an engineer who needs the data, and an executive who needs the summary.

- Primary audience: Who makes the decision based on this document?
- Secondary audience: Who needs the technical detail to implement it?
- Tertiary audience: Who needs to be informed but won't act on it?

> 📌 **ANNOTATION:** *The executive summary exists because different audiences read different sections. Design your document so each audience can navigate to what they need without reading everything.*

4. Evidence Before Conclusion

In technical writing, conclusions follow evidence. State your finding, then support it. Never lead with interpretation before data.

> **DO THIS:** Load testing revealed a 23% strength reduction at joints. We recommend redesigning the joint assembly before production.

> **NOT THIS:** We recommend redesigning the joint assembly because our testing showed some issues with the joints.

Style Rules

- Numbers: Spell out zero through nine. Use numerals for 10 and above, and for all measurements (e.g., 3 mm, 5 kg).
- Units: Always include units. '30 degrees' means nothing. '30°C' is precise.
- Abbreviations: Define on first use. (e.g., 'finite element analysis (FEA)'). Use the abbreviation consistently afterward.
- Headings: Use them liberally. Readers scan before they read. Headings are navigation tools.
- Sentence length: Vary it. Technical documents should not read like a list of facts. Rhythm matters.
- Paragraphs: One idea per paragraph. If you can't summarize a paragraph in one sentence, it's two paragraphs.
- Tables and figures: Every table and figure needs a number, a caption, and a reference in the body text before it appears.

> **COMMON ETCS WRITING ERRORS:** Undefined abbreviations. Missing units. Vague verbs (shows, indicates, demonstrates — what specifically does it show?). Passive voice hiding the actor. Data presented without interpretation. Graphics with no captions or body text reference.

Component 2

Proposal Template

Used in: ENGL 23401 and ENGL 42101

A proposal argues that a specific problem exists, a specific solution is feasible, and a specific team is qualified to execute it. Every section serves this argument.

Cover Information

[REQUIRED] Project Title: *A specific, descriptive title. Not 'Senior Project Proposal' — what is it actually?*

[REQUIRED] Submitted to: *Name, title, organization*

[REQUIRED] Submitted by: *Full name(s), student ID(s), course section*

[REQUIRED] Date: *Month Day, Year*

[REQUIRED] Course: *ENGL 23401 or ENGL 42101*

> 📌 **ANNOTATION:** *The cover page signals professionalism before the reader opens the document. Inconsistent formatting, missing fields, or typos here create doubt about the quality of the work inside.*

Section 1: Executive Summary

Executive Summary (150–200 words)

Summarize the problem, proposed solution, key deliverables, timeline, and budget in one paragraph. This section is written last but appears first.

> 📌 **ANNOTATION:** *The executive summary is the most important section. Decision-makers often read only this. It must be complete enough to stand alone. Write it after you've written everything else.*

Section 2: Problem Statement

Problem Statement

Describe the problem with specificity. What is it? Who does it affect? What evidence exists that it is a real problem? What happens if it is not solved?

> 📌 **ANNOTATION:** *Weak proposals describe vague problems. A strong problem statement includes data, stakeholders affected, and consequences of inaction. If you cannot quantify the problem, you have not researched it enough.*

Section 3: Proposed Solution

Proposed Solution

Describe your approach. What will you do? How does it address the problem? What makes this solution feasible? Consider addressing alternative solutions you considered and why you chose this one.

> 📌 **ANNOTATION:** *The proposed solution must be logically connected to the problem statement. Readers will notice if you describe one problem and propose a solution to a different one.*

Section 4: Scope & Deliverables

Scope

What is included in this project? What is explicitly out of scope? Be specific about what you will and will not deliver.

Deliverables

List each deliverable with a brief description. What will exist at the end of this project that does not exist now?

> 📌 **ANNOTATION:** *Scope defines the contract. A clear scope protects you from scope creep and protects your reader from false expectations. 'We will design a prototype' is vague. 'We will deliver a functional prototype capable of lifting 5 kg at 2 m/s' is a deliverable.*

Section 5: Timeline

Timeline / Schedule

Present a schedule with milestones and deadlines. A table or Gantt chart is preferred. Include who is responsible for each milestone.

> 📌 **ANNOTATION:** *Reviewers look at timelines to assess feasibility. A timeline that has every deliverable due in the final week signals poor planning. Build in review time, revision time, and contingency.*

Section 6: Budget

Budget

Itemize costs: materials, equipment, labor, overhead, contingency. Include a total. If your project has no budget, state that explicitly and explain why.

> 📌 **ANNOTATION:** *Even if budget is not a constraint in your course, practicing budget documentation is essential. Proposals without budgets are not taken seriously in professional contexts.*

Section 7: Qualifications

Team Qualifications

Why is your team capable of executing this project? What relevant skills, experience, or coursework do you bring? What resources do you have access to?

> 📌 **ANNOTATION:** *This section answers the reader's implicit question: 'Why should I trust you to do this?' Be specific. 'We have experience in programming' is weak. 'Two team members completed ENGR 301 and have 200+ hours of Python experience' is credible.*

Section 8: References

References

Cite all sources using the style your instructor or discipline specifies (IEEE, APA, etc.). Every claim in the proposal that is not your own original finding needs a citation.

Component 3

Technical Report Template

Used in: ENGL 23401 and ENGL 42101

A technical report documents a process, test, or investigation and communicates findings to a specific audience. It is not a lab report. It is a professional communication document.

Front Matter

[REQUIRED] Title: *Specific, descriptive. Include the subject and the type of report.*

[REQUIRED] Prepared for: *Audience name, title, organization*

[REQUIRED] Prepared by: *Author name(s), credentials, organization*

[OPTIONAL] Report Number: *If applicable*

[REQUIRED] Date: *Month Day, Year*

Section 1: Abstract (150–250 words)

Abstract

Summarize the purpose, methods, key findings, and conclusions. The abstract stands alone — a reader should understand the report's value without reading further.

> 📌 **ANNOTATION:** *The abstract is not an introduction. It is a compression of the entire report. Write it last. Include numbers and specific findings, not vague statements like 'results were analyzed and conclusions were drawn.'*

Section 2: Introduction

Introduction

State the purpose of the report. Provide necessary background. Define scope and limitations. Preview the report structure.

> 📌 **ANNOTATION:** *The introduction answers: What is this? Why does it matter? What will I find in this document? It is not a summary — that's the abstract. It is an orientation.*

Section 3: Background/Literature Review

Background

What does the reader need to know to understand your work? What prior research or standards are relevant? Where does your work fit in the larger context?

Section 4: Methods/Procedures

Methods

Describe what you did in enough detail that a qualified colleague could replicate it. Use past tense. Organize chronologically or by process. Include equipment, materials, and conditions.

> 📌 **ANNOTATION:** *Methods sections are often written too vaguely. 'We tested the samples' is insufficient. 'We subjected three 50mm×50mm aluminum samples to tensile loads of 0–5 kN in 0.5 kN increments at 22°C' is replicable.*

Section 5: Results

Results

Present data clearly and objectively. Use tables, figures, and charts. Do not interpret results here — just present them. Every figure and table must be referenced in the text before it appears.

> 📌 **ANNOTATION:** *Results and Discussion are separate sections for a reason. Results present data. Discussion interprets it. Mixing them confuses readers and makes the document harder to skim.*

Section 6: Discussion

Discussion

Interpret your results. What do they mean? How do they answer your research question? What are the implications? What are the limitations of your findings?

Section 7: Conclusions & Recommendations

Conclusions

Summarize key findings. State conclusions clearly and directly.

Recommendations

What should the reader do based on your findings? Be specific and actionable.

> 📌 **ANNOTATION:** *Conclusions answer: What did we learn? Recommendations answer: What should we do? These are different questions. Many student reports combine them badly. Keep them conceptually distinct even if they appear in the same section.*

Back Matter

References

All sources cited in the report, formatted per your discipline's style guide.

Appendices

Supporting material too detailed for the body: raw data, calculations, detailed schematics, extended code samples.

Component 4

Executive Summary Template

Used in: ENGL 23401 and ENGL 42101

An executive summary is a standalone document that distills a longer report for a decision-making audience. It is not an introduction. It is a compression.

> 📌 **ANNOTATION:** *Executive summaries are the most read and least well-written section of most technical documents. Decision-makers read them in 90 seconds. Every sentence must earn its place.*

Standard Structure (1–2 pages maximum)

[REQUIRED] Document Title: *Title of the full report this summarizes*

[REQUIRED] Purpose: *One sentence: What is this report and why does it exist?*

Problem / Context (2–3 sentences)

What problem or question prompted this work? Why does it matter?

Methods / Approach (1–2 sentences)

How was the work conducted? At a high level only.

Key Findings (3–5 bullet points)

The most important results. Include numbers. Be specific.

Conclusions (2–3 sentences)

What do the findings mean? What has been determined?

Recommendations (2–4 bullet points)

What should the reader do? Actionable, specific, prioritized.

Next Steps / Timeline (optional)

If relevant: what happens next and when?

> **THE 5-SENTENCE TEST:** Read your executive summary. Could you cut it to 5 sentences and still convey the essential information? If not, it may have too much detail. Executive summaries lose their value when they become mini-reports.

> **WHAT NOT TO INCLUDE:** Background that only specialists need. Raw data or methodology detail. Jargon without definition. Hedging language (‘it appears that,’ ‘it may be possible that’). Conclusions that don’t appear in the full report.

Component 5

Progress Report Template

Used in: ENGL 23401 and ENGL 42101

A progress report tells a supervisor or client what has been accomplished, what is in progress, what problems have arisen, and what comes next. It is a communication tool, not a diary.

Header

[REQUIRED] To: *Supervisor / Client name and title*

[REQUIRED] From: *Your name(s)*

[REQUIRED] Date: *Month Day, Year*

[REQUIRED] Subject: *Progress Report: [Project Name], [Reporting Period]*

[OPTIONAL] Report Number: *If part of a series (e.g., Progress Report 2 of 4)*

Section 1: Summary

Summary (3–5 sentences)

Overall project status. Is the project on track? What is the most important thing the reader should know?

> 📌 **ANNOTATION:** *The summary is the first thing readers see after the header. It should tell them whether to be concerned. A project that is on track, behind schedule, or facing a critical decision should be immediately clear.*

Section 2: Work Completed

Completed Since Last Report

What tasks have been finished? Use specific deliverables, not activities. 'Completed circuit board schematic' not 'worked on schematics.'

Section 3: Work in Progress

Current Activities

What is actively underway? What is the current status of each task? When will each be complete?

Section 4: Problems & Solutions

Problems Encountered

What obstacles, delays, or unexpected issues have arisen? Be direct. Do not minimize problems.

Proposed Solutions

What are you doing to address each problem? What do you need from the reader?

> 📌 **ANNOTATION:** *This section is where many student writers fail. They minimize problems or omit them entirely. Professional communication requires transparency. Your supervisor cannot help you solve a problem they don't know exists.*

Section 5: Next Steps

Planned Work

What will be accomplished before the next report? List tasks with assigned owners and target dates.

Section 6: Updated Schedule (if applicable)

Schedule Update

If the timeline has changed, show the revised schedule. Explain any changes. Do not simply present a new schedule without acknowledging what changed and why.

Component 6

Recommendation Report Template

Used in: ENGL 23401 and ENGL 42101

A recommendation report evaluates two or more options against defined criteria and recommends a course of action. It is a structured argument backed by evidence.

Cover/Header

[REQUIRED] To: *Decision-maker name, title, organization*

[REQUIRED] From: *Your name(s)*

[REQUIRED] Date: *Month Day, Year*

[REQUIRED] Subject: *Recommendation: [What is being recommended and for what purpose?]*

Section 1: Purpose & Background

Purpose

Why does this decision need to be made? What is the context?

Scope

What options are being evaluated? What is excluded from this analysis?

Section 2: Evaluation Criteria

Criteria

List and define each criterion used to evaluate options. Explain why each criterion matters. If criteria are weighted, explain the weighting rationale.

> 📌 **ANNOTATION:** *Criteria must be defined before options are evaluated. This prevents the common error of reverse-engineering criteria to justify a predetermined conclusion. Define criteria first, then apply them.*

Section 3: Options Analysis

Option A: [Name]

Describe the option. Evaluate it against each criterion. Present data or evidence. Be objective.

Option B: [Name]

Describe the option. Evaluate it against each criterion. Present data or evidence. Be objective.

Option C (if applicable): [Name]

Additional options follow the same pattern.

> 📌 **ANNOTATION:** *Evaluate each option against the same criteria in the same order. This makes comparison easy for the reader and demonstrates systematic thinking.*

Section 4: Comparison Summary

Comparison Table

A table comparing all options against all criteria is highly effective here. Readers can scan it and understand the evaluation at a glance.

Section 5: Recommendation

Recommendation

State your recommendation directly and unambiguously. Explain why this option is best given the criteria. Address potential objections.

> 📌 **ANNOTATION:** *Do not hedge your recommendation. 'Option B may be a viable choice depending on circumstances' is not a recommendation. 'We recommend Option B because it meets all three criteria within the project budget and timeline' is a recommendation.*

Section 6: Implementation / Next Steps

Next Steps

What needs to happen to implement the recommendation? Who does what, and when?

Component 7

Instructions Guide Template

Used in: ENGL 23401

Instructions tell a specific user how to complete a specific task safely and correctly. They are not descriptions. They are directives.

Front Matter

[REQUIRED] Title: *Instructions for [Specific Task]. Not 'User Manual' — what task specifically?*

[REQUIRED] Audience: *Who is this written for? What do they already know? What do they not know?*

[OPTIONAL] Version: *Date or version number*

Section 1: Introduction

Purpose

What will the user be able to do after following these instructions?

Scope

What does this guide cover? What does it not cover?

Prerequisites

What must the user have, know, or have done before starting?

Section 2: Safety Warnings & Cautions

Safety Information

List all warnings (danger to person), cautions (danger to equipment), and notes (important information). Use standard formatting: WARNING, CAUTION, NOTE in bold. Place warnings BEFORE the step they apply to, not after.

> 📌 **ANNOTATION:** *This sequencing rule is critical. A warning that appears after the dangerous step is useless. Always place safety information before the action the reader must avoid or approach carefully.*

Section 3: Materials & Equipment

Required Materials

List everything the user needs before they start. Include part numbers, quantities, and specifications where relevant.

Section 4: Step-by-Step Instructions

Steps

Number each step. One action per step. Use imperative voice (Connect the wire… not 'The wire should be connected…'). Include expected outcomes for key steps so users can verify they are on track.

> 📌 **ANNOTATION:** *Instructions that bundle multiple actions in one step create errors. 'Attach the bracket and tighten the bolts and check alignment' is three steps. Write them as three steps.*

Section 5: Troubleshooting

Common Problems & Solutions

What can go wrong? Present as: Symptom → Possible Cause → Solution. Cover the top 5–7 most common issues.

Section 6: Glossary (if applicable)

Glossary

Define technical terms that may be unfamiliar to your target audience.

Component 8

White Paper Template

Used in: ENGL 42101

A white paper presents a problem, explains its causes, and proposes or advocates for a solution. It is longer and more developed than a recommendation report, typically aimed at an industry or policy audience.

> 📌 **ANNOTATION:** *White papers are persuasive documents grounded in evidence. They are not academic essays and not sales documents. They occupy the space between: expert analysis that advocates for a position.*

Cover Page

[REQUIRED] Title: *The title should name the problem and/or solution clearly. 'Reducing Thermal Failure in High-Density PCB Design' not 'PCB White Paper.'*

[REQUIRED] Authors: *Name(s), title(s), affiliation(s)*

[REQUIRED] Date: *Month Year*

[OPTIONAL] Organization: *Sponsoring organization or course context*

Section 1: Abstract (200 words)

Abstract

Problem, approach, key finding, recommendation. Self-contained.

Section 2: Introduction

Introduction

What problem does this white paper address? Why does it matter now? Who should read this? What will the reader gain?

Section 3: Problem Definition

Problem Definition

Define the problem with specificity and evidence. What causes it? What are its consequences? What is the current state of practice and why is it insufficient?

> 📌 **ANNOTATION:** *The depth of your problem definition determines the credibility of your solution. A shallow problem statement produces an unpersuasive recommendation. Invest the most space and evidence here.*

Section 4: Background & Context

Context

What has been tried before? What does the literature or industry practice say? Where do current approaches fall short?

Section 5: Proposed Solution / Position

Solution

What do you recommend? What evidence supports this approach? How does it address the problem as you have defined it?

Section 6: Implementation Considerations

Implementation

What would it take to adopt this solution? What are the barriers? What resources are required? What are the risks?

Section 7: Conclusion

Conclusion

Restate the problem. Restate the solution. Call to action: what should the reader do next?

References

References

White papers require rigorous citation. Every factual claim needs a source. Use the style appropriate to your field.

Component 9

General Usability Guide

Used in: ENGL 23401 and ENGL 42101

Usability is not about making documents pretty. It is about making documents navigable, scannable, and readable for a specific audience under real conditions.

Document-Level Usability

- Every document needs a title that tells the reader what it is and what it is for.
- Long documents (5+ pages) need a table of contents.
- Section headers should be descriptive, not generic. 'Test Results' is generic. 'Tensile Strength Results at 22°C' is descriptive.
- Page numbers are required on all multi-page documents.
- Consistent formatting signals credibility. Inconsistent formatting signals carelessness.

Page-Level Usability

- White space is functional, not wasted. Dense pages discourage reading.
- Margins should be at least 1 inch on all sides.
- Body text should be 11 or 12pt in a readable font (Calibri, Times New Roman, Arial).
- Line spacing of 1.15 to 1.5 improves readability for dense technical content.
- Left-aligned body text is easier to read than justified text for most readers.

Heading Hierarchy

Use a consistent heading hierarchy throughout the document:

> **H1 (MAJOR SECTIONS):** 16–18pt, bold, navy or dark color — used for Introduction, Methods, Results, Conclusions
>
> **H2 (SUBSECTIONS):** 13–14pt, bold — used for subdivisions within major sections
>
> **H3 (SUB-SUBSECTIONS):** 12pt, bold or italic — used sparingly for detailed breakdowns

> 📌 **ANNOTATION:** *Never use a heading level that skips a level (e.g., H1 directly to H3). This confuses the document's hierarchy and signals poor organization.*

Lists & Tables

- Use bullet lists for unordered items where sequence doesn't matter.
- Use numbered lists for sequential steps or ranked items.
- Tables are preferred over text for comparative data with three or more variables.
- Every table needs a table number and a descriptive caption above the table.
- Every figure needs a figure number and a descriptive caption below the figure.

Accessibility Checklist

- Color alone is never used to convey meaning (for colorblind readers).
- All images have descriptive alt text or figure captions.
- Font size is at least 11pt for body text.
- Contrast between text and background is sufficient (dark text on light background or vice versa).
- Abbreviations are defined on first use.
- The document reads in logical order without relying on visual layout alone.

Component 10

Graphics & Visuals Guidance

Used in: ENGL 23401 and ENGL 42101

Graphics are not decoration. They are communication tools. Every graphic in a technical document must earn its place.

The Three Rules of Technical Graphics

1. Every graphic must be referenced in the body text before it appears. (e.g., 'Figure 3 shows the stress distribution across the joint.')
2. Every graphic must have a number and a descriptive caption. (e.g., 'Figure 3. Von Mises stress distribution, aluminum joint at 4 kN load.')
3. Every graphic must add information the text cannot convey efficiently. If you can explain it in one sentence, you probably don't need the graphic.

Choosing the Right Visual

> **LINE CHARTS:** Trends over time or continuous data. Best for showing how a variable changes.
>
> **BAR CHARTS:** Comparing discrete categories. Best for side-by-side comparison of values.
>
> **TABLES:** Comparing multiple variables across multiple items. Best when exact values matter.
>
> **SCHEMATICS / DIAGRAMS:** Explaining how something works or is structured. Best for spatial or process information.
>
> **PHOTOGRAPHS:** Showing physical reality. Best for documenting conditions, equipment, or results that must be seen.

> **AVOID PIE CHARTS:** Pie charts are difficult to read accurately and rarely the best choice for technical data. Use bar charts instead.

Graphic Design Principles for Technical Documents

- Label axes, units, and data series on all charts.
- Use consistent color coding across all graphics in a document.
- Avoid decorative 3D effects on charts — they distort data perception.
- Scale graphics consistently when comparing across the document.
- Minimum resolution for print is 300 DPI. For digital documents, 150 DPI is acceptable.
- If a graphic is from another source, it requires a citation in the caption.

> **THE 'SO WHAT' TEST:** After every graphic, ask: So what? If you cannot answer that question in one sentence, the graphic may not be communicating effectively. Add an interpretive sentence in the body text that explains what the reader should take away from the graphic.

Component 11

Annotated Sample: Engineering Proposal

Used in: ENGL 23401 and ENGL 42101

This sample demonstrates a well-structured student engineering proposal. Annotations explain the design decisions behind each section. Use this as a model, not a template to fill in verbatim. NOTE: This is a work of fiction.

Automated Campus Irrigation System

A Proposal for Water-Efficient Landscaping at Hartfield University

Submitted to: Dr. Elizabeth Keller Kirycki

Submitted by: J. Torres, K. Chen, P. Osei | ENGL 42101 | March 14, 2026

> 📌 **ANNOTATION:** *The title is specific and descriptive. It names the deliverable (automated irrigation system), the context (campus), and the purpose (water-efficient landscaping). The subtitle names the recipient and location. The reader knows immediately what this document is and for whom.*

Executive Summary

Hartfield University's manual irrigation system consumes approximately 2.3 million gallons of water annually, 34% more than comparable campuses using automated systems (EPA WaterSense, 2024). This proposal outlines a sensor-based automated irrigation system for the university's 12-acre grounds that would reduce water consumption by an estimated 25–30%, yielding annual savings of $18,000–$22,000 at current municipal rates. The system can be installed in two phases over 14 months for a total estimated cost of $74,500, with projected full cost recovery within 3.5 years. We recommend approval of Phase 1 funding ($38,000) to begin installation of the core sensor network in the east and central campus zones.

> 📌 **ANNOTATION:** *This executive summary does everything right: specific problem (2.3M gallons, 34% above comparable), specific solution (sensor-based automated system), quantified benefit (25–30% reduction, $18K–$22K savings), phased budget ($38K Phase 1), and a clear ask (approval of Phase 1). A decision-maker reading only this paragraph has what they need.*

Problem Statement

Hartfield University's 12-acre grounds are currently irrigated using a manual timer-based system installed in 2008. The system operates on fixed schedules regardless of rainfall, soil moisture, or seasonal variation. As a result, the university irrigates during and after rain events, overwatering turf and ornamental beds an estimated 30–40% of the time (Facilities Management Water Audit, 2025).

This inefficiency has three consequences: (1) unnecessary water expenditure averaging $22,000 annually above what an automated system would require; (2) soil compaction and turf damage from overwatering in high-traffic zones; and (3) non-compliance with the state's 2024 commercial irrigation efficiency guidelines, which require institutions to reduce outdoor water use by 20% by 2027.

> 📌 **ANNOTATION:** *The problem statement is grounded in three types of evidence: internal audit data, financial quantification, and external regulatory pressure. It names three specific consequences, making it clear that this is not a marginal problem. The 2027 compliance deadline creates urgency without being manipulative.*

Proposed Solution

We propose installing a smart irrigation system using soil moisture sensors, weather API integration, and zone-specific controllers across the campus grounds. The system will irrigate only when soil moisture falls below defined thresholds and will defer or cancel scheduled watering when rainfall is detected or forecast.

We evaluated three approaches: (1) full replacement of the existing system, (2) overlay of smart controllers on the existing infrastructure, and (3) phased installation prioritizing high-use zones. Option 2 was eliminated due to compatibility issues with the 2008 valves. We recommend Option 3 as the most cost-effective approach that achieves compliance within the 2027 deadline.

> 📌 **ANNOTATION:** *The solution section explains what will be done, shows that alternatives were considered and why they were rejected, and justifies the recommended approach. This demonstrates systematic thinking and builds confidence in the recommendation.*